

DO VAN KHA

BACKEND DEVELOPER / SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Backend Developer and Software Engineer focused on System Architecture & Optimization with practical AI Integration. Passionate about engineering sustainable, high-performance backends, clean code standards, and robust containerization (Docker). Experienced in designing resilient APIs, microservices, and AI-powered pipelines (RAG & AI Agents).

TECHNICAL SKILLS

Backend & Languages: TypeScript, JavaScript (ES6+), Node.js, NestJS, Express.js

Databases & Caching: PostgreSQL, MongoDB, Redis, MySQL, Database Optimization

DevOps & Cloud: Docker, Git / GitHub, Vercel, Railway, CI/CD Pipelines

AI & Modern Tools: Retrieval-Augmented Generation (RAG), AI Agent Integration, LangChain, Vitest, Swagger, Postman

PROJECTS & EXPERIENCE

SmartQuery — Intelligent AI Query Engine | 2026

Role: System Architecture & Backend Engineer

Tech Stack: Node.js, MySQL, AI Agents Integration, Docker

- Architected a robust Prompt & Context processing pipeline enabling AI Agents to translate natural language into accurate, optimized SQL queries.
- Engineered secure connection pooling and multi-layered defense mechanisms against SQL Injection from AI-generated outputs.
- Optimized database schema context ingestion to provide LLMs with precise structural relationships for high accuracy query generation.
- Containerized full application ecosystem using Docker for seamless development and deployment pipelines.

Backend Engineering & System Optimization | Ongoing

Focus: High-Performance Distributed Systems

- Designing modular backend architectures with NestJS implementing dependency injection and enterprise design patterns.
- Implementing distributed caching strategies with Redis to minimize latency and optimize query throughput.
- Exploring message queue architectures (RabbitMQ/Kafka) for event-driven asynchronous services.

EDUCATION

Software Engineering | Ongoing

Undergraduate Degree in Software Engineering

- Focus on Data Structures, Algorithms, Database Management Systems, and Distributed Computing.